

FINAL SCHEME

02000MAT206052104

Total Pages: 4			
Scheme of Valuation/Answer Key			
(Scheme of evaluation (marks in brackets) and answers of problems/key)			
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY			
B.Tech Degree S4 (R,S) / S2 (PT) (R,S) Examination June 2023 (2019 Scheme)			
Course Code: MAT206			
Course Name: GRAPH THEORY			
Max. Marks: 100			Duration: 3 Hours
PART A			
		(Answer all questions; each question carries 3 marks)	Marks
1		walk.....(1 Mark) Path.....(1 Mark) Circuit.....(1 Mark)	3
2		A graph G is connected if there exist at least one path between every pair of vertices in G.....(2 Marks) Correct example.....(1 Mark)	3
3		Since the degree of each vertex is even, the given graph is Euler.....(1 Mark) Euler tour is given by $v_1e_1v_2e_6v_5e_{10}v_6e_{11}v_7e_9v_5e_7v_8e_8v_7e_{12}v_3e_4v_5e_5v_2e_2v_3e_3v_1$(2 Marks), Any other correct Euler tour (2 Marks)	3
4		An equivalence digraph is a directed graph which represents an Equivalence relation. It should be reflexive, symmetric and transitive.....(2 Marks) Correct example.....(1 Mark)	3
5		A vertex in a graph G with minimum eccentricity is called center of G.....(1 mark) $E(v_1)=2$, $E(v_2)=1$, $E(v_3)=2$ and $E(v_4)=2$(1 Mark) Center of G is v_2(1 Mark)	3
6		For correct proof.....(3 Marks)	3
7		$S_1=\{e_1,e_6\}$ $S_2=\{e_3,e_4\}$ $S_3=\{e_1,e_7,e_5\}$ $S_4=\{e_2,e_3,e_9\}$ $S_5=\{e_2,e_6,e_7,e_8\}$(2 Marks)	3

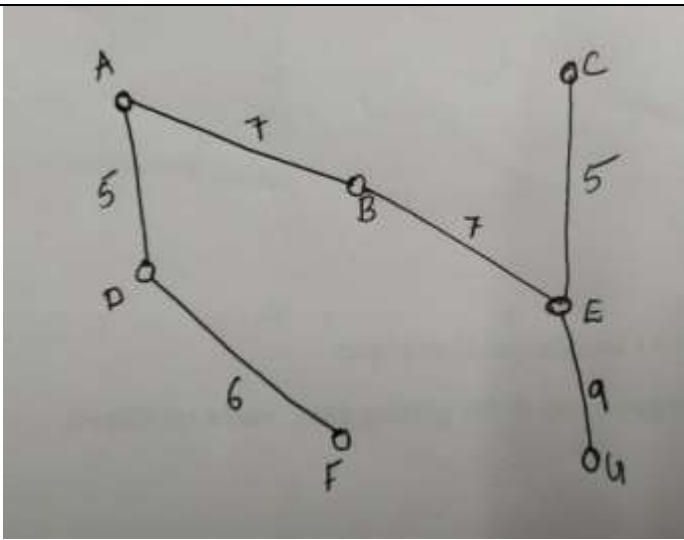
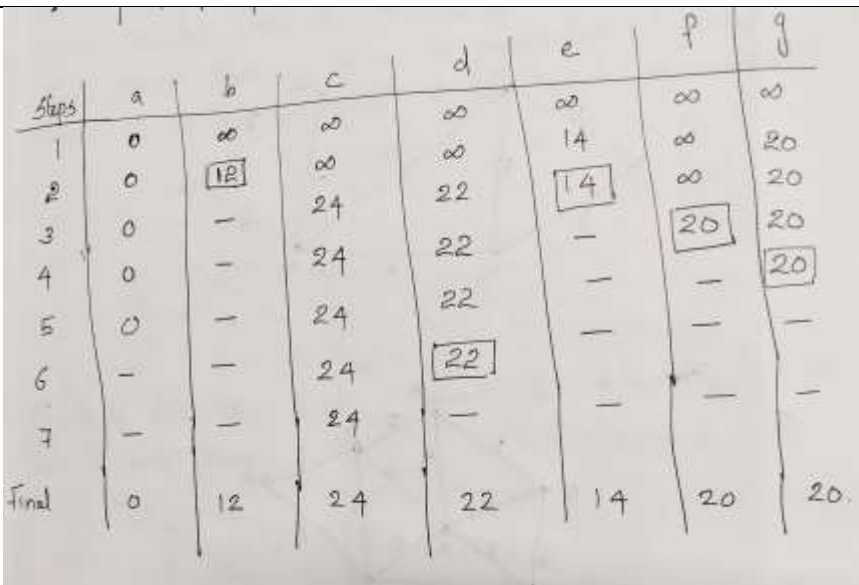
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		Edge connectivity is 2.....(1 Mark)	
8		Drawing $K_{3,3}$(1.5 Marks) Drawing K_5(1.5 Marks) Drawing any two non planer graph, give full marks	3
9		$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$(3 Marks)	3
10		The minimum no. of colours required to colour a graph properly is called the chromatic number. /The minimum no. of colours required to colour a graph so that no two adjacent vertices receive the same colour is called its chromatic number.....(2 Marks) Example.....(1 Mark)	3
PART B			
<i>(Answer one full question from each module, each question carries 14 marks)</i>			
Module -1			
11	a)	For correct proof.....(7 Marks)	7
	b)	For showing the one –one correspondence between vertex set and edge set of two graphs.....(5 Marks) Showing two graphs are isomorphic.....(2 Marks)	7
12	a)	For correct proof.....(8 Marks)	8
	b)	Sub graph: A graph H is said to be a sub graph of a graph G if all the vertices and all the edges of H are in G, and each edge of H has the same end vertices in H as in G(1 Mark) Edge disjoint sub graph: Two sub graphs are said to be edge disjoint if they do not have any edges in common.....(1 Mark) Vertex disjoint sub graph: Two sub graphs are said to be vertex disjoint if they do not have any vertices in common.....(1 Mark) Drawing 3 examples.....(3 Marks)	6
Module -2			
13	a)	Necessary condition - 4 marks Sufficient condition - 4	8

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	b)	Symmetric digraph- A digraph in which for every edge (a,b)there is also an edge (b,a).....(2 Marks) Asymmetric digraph- Digraphs that have atmost one directed edge between a pair of vertices, but are allowed to have loops.....(2 Marks) Drawing example of both symmetric and asymmetric digraph.....(2 Marks)	6
14	a)	Travelling salesman problem-correct explanation.....(7 Marks)	7
	b)	For correct proof.....(7 Marks)	7
Module -3			
15	a)	For correct proof.....(7 Marks)	7
	b)	 Give suitable stage marks- 7 marks	7
16	a)	For correct proof.....(6 Marks)	6
	b)	 (8 Marks)	8

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Module -4			
17	a)	Statement- If G is a connected plane graph, then $n-m+f= 2$ where n,m,f denotes number of vertices, number of edges and number of faces respectively.....(2 Marks) For correct proof.....(6 Marks)	8
	b)	Vertex connectivity-Let G be a graph. The minimum number of vertices whose deletion from G increase the number of components of G is called the vertex connectivity.....(2 Marks) Edge Connectivity- Let G be a graph. The minimum number of edges whose deletion from G increase the number of components of G is called the edge connectivity.....(2 Marks) For drawing correct example.....(2 Marks)	6
18	a)	For correct proof.....(7 Marks)	7
	b)	For correct proof.....(7 Marks)	7
Module -5			
19	a)	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 \end{bmatrix}$(5 Marks)	5
	b)	For correct proof.....(9 Marks)	9
20	a)	Circuit Matrix- Let G be a graph with e edges and q different cycles. The circuit matrix $B(G)=[b_{i,j}]$ of order $q \times e$ such that $b_{i,j} = \begin{cases} 1 & \text{if the } i\text{th circuit includes } j\text{th edge} \\ 0; & \text{other wise} \end{cases}$(2 Marks) Circuit matrix is $C(G) = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \end{bmatrix}$(4 Marks)	6
	b)	For correct proof.....(8 Marks)	8
